

# Learning Joint Text and Visual Tokens in CLIP for Medical Image Analysis

Raffaele Mineo\*  
University of Catania  
Catania, Italy  
raffaele.mineo@unict.it

Giovanni Bellitto\*  
University of Catania  
Catania, Italy  
giovanni.bellitto@unict.it

Federica Proietto Salanitri  
University of Catania  
Catania, Italy  
federica.proiettosalanitri@unict.it

Rutger Hendrix  
University of Catania  
Catania, Italy  
rutger.hendrix@phd.unict.it

Concetto Spampinato  
University of Catania  
Catania, Italy  
concetto.spampinato@unict.it

Simone Palazzo  
University of Catania  
Catania, Italy  
simone.palazzo@unict.it

## Abstract

We present TVTL-CLIP, a parameter efficient prompt tuning strategy that adapts a frozen CLIP model to specialized medical-image analysis without updating its millions of pretrained weights. TVTL-CLIP inserts two compact sets of learnable tokens, 64 class-specific text prompts and 5 deep visual prompts, into CLIP’s text and vision encoders, jointly steering both modalities toward domain-specific features that are absent from natural-image data. This design use only 0.08M trainable parameters ( $<0.1\%$  of ViT-B/32) yet preserves the rich general representations of the underlying model. On a capsule endoscopy benchmark of 1’842 images spanning seven gastrointestinal lesions types, TVTL-CLIP reaches 88.7% accuracy, 79.2% recall, and 98.3% specificity, surpassing state-of-the-art CNNs and vision transformers while using two to three orders of magnitude fewer learnable weights. Extensive ablations confirm that (i) visual and textual prompts are individually beneficial but deliver the largest gain when combined, and (ii) a handful of prompts per layer is sufficient, larger prompt sets overfit. The results demonstrate that prompt based, dual modality conditioning is a practical route to high accuracy, low footprint medical classifiers, making TVTL-CLIP a compelling choice for real-world deployments where annotated data and compute budgets are severely limited.

## CCS Concepts

• **Computing methodologies** → **Computer vision; Neural networks**; • **Applied computing** → **Life and medical sciences**.

## Keywords

Vision-language models, Prompt tuning, Soft prompts, Parameter efficient adaptation, Medical image analysis, Gastrointestinal pathology classification

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## 1 Introduction

Recent advances at the intersection of computer vision and natural language processing have propelled the development of large scale vision-language models, which learn joint image-text representations from vast, weakly-annotated web data. These models capture highly transferable features that can be rapidly fine-tuned, with only limited supervision, for a broad spectrum of downstream applications.

In NLP, *prompt learning* has emerged as an efficient alternative to full model fine-tuning. Early work mined or paraphrased discrete prompts[8], while AutoPrompt[18] optimised token choices via gradient signals. More recent *continuous* methods tune a small set of learnable vectors in the embedding space[11, 13, 35], achieving competitive accuracy with orders-of-magnitude fewer trainable parameters.

CLIP[16] demonstrated that contrastive pre-training on paired image-text data yields rich, generalisable features. Subsequent studies confirmed that scaling dataset size and using transformer backbones further improve performance[6, 19]. CLIP supports zero-shot recognition and serves as a backbone for more complex systems such as latent-diffusion generators[17] and segmentation models[9]. Yet, when deployed in specialised domains (e.g. medical imaging) its zero-shot accuracy drops markedly because domain-specific visual patterns and terminology deviate from the web data (mainly from natural image domain) on which CLIP was trained.

Adapting a vision transformer to a new medical task by updating all weights is computationally expensive and prone to overfitting, especially when labelled data are scarce. Domain-oriented foundation models have begun to appear[15, 20, 33]; however, their size still renders task-specific finetuning prohibitive in many clinical settings.

Prompt-based approaches such as CoOp[37] and VPT[7] require only a few additional parameters, but the medical domain introduces extra challenges: domain shift affects both the visual stream

\*Equally contributed.



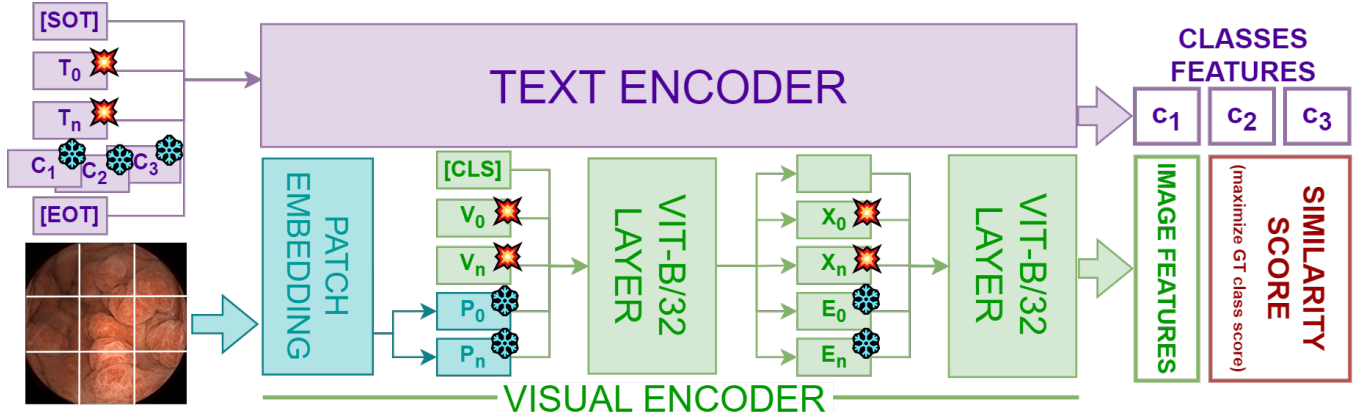
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**Figure 1: Architecture of the Proposed Approach.** At the top, the textual encoder takes learnable tokens and labels as input and generates embeddings for each class. The correct class should minimize the similarity score with the visual embedding outputted by the visual encoder (ViT-B-32), which takes learnable patches and visual patch embeddings as input.

(pathology-specific textures) and the textual stream (specialist terminology like "high-grade dysplasia in Barrett's oesophagus"). A strategy that *simultaneously* adapts *both* modalities is therefore crucial.

We propose *Text and Visual Tokens Learnable in CLIP* (TVTL-CLIP), a parameter-efficient framework that inserts distinct sets of *soft prompts* into the frozen CLIP text and vision encoders (Fig.1). The prompts steer the encoders to attend to medically relevant features without disturbing the pretrained weights. We evaluate TVTL-CLIP on 1'842 capsule endoscopy images covering seven lesion categories (angiodyplasia, bleeding, erosion, hyperaemia, petechial, polyp, ulcer). TVTL-CLIP surpasses convolutional and transformer baselines while training two orders of magnitude fewer parameters, demonstrating that joint visual-textual prompting is a practical path toward reliable, data-efficient medical image classification.

## 2 Related Work

Large-scale contrastive vision-language pre-training has produced a new generation of foundation models that align images and text in a shared latent space. Landmark systems such as CLIP[16], ALIGN[6], BLIP[12], Florence[26], CoCa[24], and SimVLM[21] are trained on hundreds of millions to billions of web scale image-caption pairs and achieve impressive zero-shot and few-shot performance on classification, retrieval, detection, and generation. Their success emerges from the combination of transformer backbones [19], massive heterogeneous data, and contrastive objectives that tightly couple the two modalities, yielding generalizable visual and linguistic features.

Adapting these models to specialized domains, however, is non trivial. Full fine-tuning entails updating hundreds of millions of weights and risks severe overfitting when labeled data are scarce. A rich body of work therefore explores parameter-efficient strategies. Adapter layers[4], BitFit[30], and LoRA[5] modify only small subsets of parameters. Prompt-based methods go further and tune a handful of context tokens: Visual Prompt Tuning (VPT)[7] inserts learnable image tokens, while CoOp[37] and CoCoOp[36] optimize

text prompts, the latter making them instance adaptive. Extensions such as DualPrompt[22], probe different insertion depths or split prompts into multiple levels, repeatedly demonstrating competitive accuracy with less than one percent of the backbone parameters. Recently, multi-modal prompt learning has been extended beyond recognition to perceptual quality assessment tasks, e.g. CLIP-AGIQA[2] for AI-generated images and CLIP-DQA[31] for dehazed image quality, where learnable prompts in both vision and language branches enable better alignment between semantic descriptions and perceptual quality cues.

Prompt learning itself originated in natural language processing. After early work with manual templates, automatic discovery methods emerged: Jiang et al.[8] mine candidate phrasings, AutoPrompt[18] chooses discrete tokens with gradient signals, and continuous approaches, like prefix tuning[13], prompt tuning[11], P-tuning v2[14], and factual probing[35], optimize embedding vectors directly, shrinking task specific storage to a few kilobytes.

Domain specific applications present additional challenges. In medical imaging, visual patterns such as subtle mucosal textures and textual labels like "Barrett's oesophagus with high grade dysplasia" are largely absent from web data, causing a pronounced domain shift that degrades zero-shot accuracy. Several efforts tackle this by pretraining on clinical data: BioViL[1], MedCLIP[33], GatorTron[23], BioMedCLIP[32], and LVM-Med[20], but their billion parameter scale still demands resource intensive fine-tuning. Moreover, hand-crafted prompts are brittle: Zhang et al.[34] show that minor wording changes (e.g. "an X-ray of ..." versus "a radiograph showing ...") can swing performance by more than twenty percentage points.

In parallel, specialized vision-language approaches have been explored for domain challenges. In colonoscopy, Yue et al.[28] introduced a large-scale video quality database with real-world distortions and a distortion-aware reasoning network, while Yue et al.[27] proposed a boundary refinement network for precise colorectal polyp segmentation. For retinal imaging, subjective evaluation benchmarks such as ERIQAD[29] quantify quality of enhanced retinal images, and autoregressive ordinal regression with diffusion priors has been proposed for diabetic retinopathy grading[25].

These works highlight the need for models that capture fine-grained medical semantics, robustly handle quality degradations, and exploit domain knowledge such as ordinal severity levels.

Finally, continual adaptation remains a pressing problem. While domain-specific pretraining partially mitigates domain shift, medical applications often require incremental incorporation of new disease categories and imaging protocols. Recent work such as CalFuse[3] leverages vision-language alignment and feature calibration to enhance class-continual learning, pointing towards hybrid strategies that integrate multimodal prompts, continual adaptation, and domain-aware quality assessment. These observations motivate methods that jointly adapt the visual and textual streams with a minimal number of learnable tokens, letting the model discover clinically salient features in both modalities while preserving efficiency.

### 3 Method

We adapt a frozen CLIP model to medical image classification by learning a *small* set of prompt tokens that steer both the vision and the text encoders toward domain-specific features while leaving all pretrained weights untouched. This strategy, illustrated in Fig. 1, combines the generalization capacity of large scale contrastive pretraining with the efficiency of parameter-light finetuning.

Let  $E_{\text{vis}}$  (a ViT-B/32) map an input image  $I$  to a visual embedding  $\mathbf{v} = E_{\text{vis}}(I)$  and  $E_{\text{text}}$  (the CLIP text transformer) map a class name  $c_n \in \mathcal{C}$  to a textual embedding  $\mathbf{t}_n = E_{\text{text}}(c_n)$ . The cosine similarity  $s_n = \langle \mathbf{v}, \mathbf{t}_n \rangle / (\|\mathbf{v}\| \|\mathbf{t}_n\|)$  serves as the compatibility score between image and label.

For each class we prepend a sequence of  $M$  learnable *soft text tokens*  $\mathbf{t}_n^{1:M} \in \mathbb{R}^d$  ( $d=512$ ) to the label, producing the prompt

$$[\text{SOT}] \mathbf{t}_n^1 \dots \mathbf{t}_n^M c_n [\text{EOT}],$$

which is encoded by the frozen  $E_{\text{text}}$ . We explore two variants: a *class-specific context* (CSC) where each class has its own tokens, and a *shared context* that reuses one pool of tokens for all classes. In either case only the prompt vectors are updated, enabling the text encoder to incorporate specialist terminology such as "*high-grade dysplasia*" without disturbing its general linguistic knowledge.

The image  $I$  is split into  $K$  patches whose embeddings  $\mathbf{p}_{1:K} \in \mathbb{R}^{d_b}$  ( $d_b=768$ ) are processed by the ViT. We insert  $M$  learnable *visual tokens*  $\mathbf{v}^{1:M}$  dimension-matched to the patch embeddings, between the CLS token  $\mathbf{cls}$  and the patch sequence; so that the input to the first transformer layer becomes

$$[\mathbf{cls}, \mathbf{v}^{1:M}, \mathbf{p}_{1:K}].$$

These tokens, shared across images, let the model focus on medically relevant visual patterns (e.g. vascular lesions) while the rest of  $E_{\text{vis}}$  stays frozen.

Given a mini-batch of  $B$  image-label pairs, we compute all pair-wise similarities  $s_{ij}$  and minimize the symmetric contrastive cross-entropy loss

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^B \left[ -\log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ij}/\tau)} - \log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ji}/\tau)} \right].$$

With  $N=8$  classes,  $M=16$  tokens and the above dimensions, CSC text prompts add  $N * M * d \approx 65k$  parameters, or only 8k when

prompts are shared. Visual prompts contribute a further  $M * d_b \approx 12k$ . In total we update fewer than 80k parameters ( $< 0.1\%$  of ViT-B/32) yet, as shown in Section 4.3, this suffices to surpass full finetuning baselines on a capsule endoscopy dataset with less than 2k images.

## 4 Experimental Results

### 4.1 Dataset

Our study leverages a proprietary dataset of gastroscopy videos acquired via capsule endoscopy between 2018 and 2021 at the University Hospital of Catania (Italy). The cohort comprises 713 patients, 317 female and 396 male, aged 43 to 79 years. In particular, each patient ingested a disposable capsule equipped with two cameras that recorded continuously at VGA resolution ( $640 \times 480$ ) and 15fps while traversing the gastrointestinal tract. From the resulting videos, two professional gastroenterologists extracted and independently annotated frames depicting a broad spectrum of gastrointestinal pathologies, producing image level labels that were reconciled by consensus.

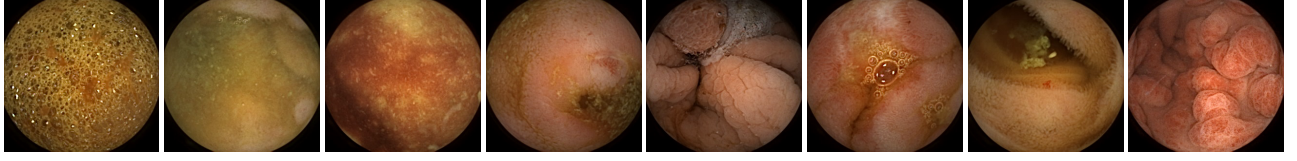
The dataset comprises a total of 1'842 images from gastroscopic examinations, with annotations categorizing them into 7 distinct pathological classes plus negative samples, as is shown in Fig. 2. Specifically, the dataset includes the following annotated pathologies:

- (1) **Negatives**: Approximately 700 images depicting normal gastroscopic findings, serving as the baseline for comparison and model evaluation;
- (2) **Erosion**: 229 images depicting erosions or superficial mucosal damage in the gastrointestinal tract;
- (3) **Bleeding**: 56 images showing evidence of gastrointestinal bleeding, including acute bleeding or blood clots;
- (4) **Petechial**: 47 images showing small, pinpoint hemorrhages in the mucous membranes of the gastrointestinal tract;
- (5) **Polyp**: 67 images depicting abnormal tissue growths protruding from the gastrointestinal mucosa;
- (6) **Ulcer**: 76 images depicting ulcers, characterized by areas of mucosal discontinuity and tissue loss in the gastrointestinal tract;
- (7) **Angiodysplasia**: 490 images depicting vascular malformations in the gastrointestinal tract;
- (8) **Hyperemia**: 132 images depicting increased blood flow and redness in the gastrointestinal mucosa, indicative of inflammation.

Each image is extracted from videos in the dataset and labeled with the corresponding pathology, enabling precise supervision during model training and evaluation. Despite the highly imbalanced distribution of classes in our dataset, our proposed methodology achieves good performance across all gastrointestinal pathologies used, including those with lower frequencies. This underscores the robustness and efficacy of our approach in clinical settings.

### 4.2 Training and Evaluation Procedure

The training methodology for the TVTL-CLIP model was regulated to optimize performance trade-off between sensitivity and specificity of gastrointestinal pathology classification tasks while ensuring computational efficiency and scalability.



**Figure 2: An example image for each class of our dataset (from left to right): *none*, *erosion*, *bleeding*, *petechial*, *polyp*, *ulcer*, *angiodysplasia*, *hyperemia*.**

In order to facilitate efficient optimization of transformer encoders, we have to use the stochastic gradient descent (SGD) optimizer. Also, to reduce the model memory footprint without visible degradation of accuracy, we use 16-bit floating point training using CUDA Automatic Mixed Precision (AMP). This approach not only accelerates training but also conserves computational resources, enabling faster convergence and model refinement. The learning rate chosen is 0.002 over 200 epochs, maintaining a balance between rapid convergence and stability throughout the training process.

One of the principal elements of our training methodology was the investigation of diverse token configurations for both the textual and visual encoders. With regard to the textual encoder, we try different numbers of learnable textual tokens, ranging from 0 to 69 (maximum permitted for pretrained text encoder, considering special and class names token), in both CSC and non-CSC modality.

**Table 1: Performance comparison with different combinations of textual and visual tokens. *Text Tok.* indicates the number of textual tokens used, *CSC* indicates whether a class-specific context was used, *Vis. Tok.* indicates the number of visual tokens added, *Deep Vis. Tok.* indicates whether visual tokens were added at each visual transformation layer.**

| Text Tok. | CSC | Vis. Tok. | Deep Vis. Tok. | Acc. | Recall | Spec. | F1 Score |
|-----------|-----|-----------|----------------|------|--------|-------|----------|
| 0         |     | 0         |                | 48.2 | 10.6   | 85.8  | 10.2     |
| 0         |     | 1         | ✗              | 50.9 | 5.9    | 95.8  | 6.8      |
| 0         |     | 1         | ✓              | 65.4 | 33.5   | 97.3  | 40.1     |
| 0         |     | 5         | ✗              | 51.6 | 6.0    | 97.3  | 6.5      |
| 0         |     | 5         | ✓              | 84.2 | 70.8   | 97.6  | 77.9     |
| 1         | ✗   | 0         |                | 54.6 | 11.5   | 97.7  | 17.6     |
| 1         | ✓   | 0         |                | 54.6 | 11.5   | 97.7  | 17.6     |
| 1         | ✗   | 1         | ✗              | 53.3 | 9.9    | 96.7  | 14.3     |
| 1         | ✗   | 1         | ✓              | 72.7 | 48.6   | 96.8  | 56.3     |
| 1         | ✓   | 1         | ✗              | 53.1 | 9.7    | 96.6  | 13.9     |
| 1         | ✓   | 1         | ✓              | 72.7 | 48.6   | 96.8  | 56.3     |
| 1         | ✗   | 5         | ✗              | 52.0 | 7.6    | 96.4  | 7.2      |
| 1         | ✗   | 5         | ✓              | 83.5 | 68.7   | 98.3  | 73.8     |
| 1         | ✓   | 5         | ✗              | 52.0 | 7.6    | 96.4  | 7.2      |
| 1         | ✓   | 5         | ✓              | 86.0 | 73.7   | 98.3  | 75.7     |
| 64        | ✗   | 0         |                | 76.9 | 57.6   | 96.2  | 68.4     |
| 64        | ✓   | 0         |                | 76.9 | 57.6   | 96.2  | 68.4     |
| 64        | ✗   | 1         | ✗              | 79.0 | 60.2   | 97.8  | 67.4     |
| 64        | ✗   | 1         | ✓              | 80.4 | 63.3   | 97.6  | 65.4     |
| 64        | ✓   | 1         | ✗              | 79.0 | 60.2   | 97.8  | 67.4     |
| 64        | ✓   | 1         | ✓              | 80.4 | 63.3   | 97.6  | 65.4     |
| 64        | ✗   | 5         | ✗              | 78.6 | 60.0   | 97.2  | 66.0     |
| 64        | ✗   | 5         | ✓              | 86.8 | 75.4   | 98.1  | 78.2     |
| 64        | ✓   | 5         | ✗              | 76.7 | 55.5   | 97.9  | 68.2     |
| 64        | ✓   | 5         | ✓              | 88.7 | 79.2   | 98.3  | 80.2     |

Similarly, the same text encoder experiments were conducted for the visual encoder, with different numbers of visual tokens, ranging

from 0 to 10, added at the input level or at each of the 12 levels of the visual encoder.

### 4.3 Results

In our experimental evaluation, we conduct an extensive grid search encompassing 156 combinations of textual and visual token configurations. These included variations of CSC/non-CSC text token and first/every layer (deep) visual token parameters. Through this exhaustive search, we identified the optimal configuration for the TVTL-CLIP model.

The efficacy of leveraging the learnable tokens of both CLIP’s textual encoder and visual encoder has been demonstrated through a series of experiments involving different quantities for textual and visual tokens. Configurations with multiple textual tokens and a single visual token consistently outperformed their counterparts without learnable token configurations. In particular, focusing on the single visual token configuration highlights the necessity of adding visual tokens to textual ones. On the other hand, incorporating multiple visual tokens without corresponding textual tokens led to overfitting, indicating the importance of balancing the presence of learnable tokens in both modalities.

Moreover, the inclusion of class-specific context (CSC) and an excessive number of visual tokens (greater than 5) resulted in performance degradation due to the increased risk of overfitting. This phenomenon underscores the delicate balance required in model parameterization to prevent overfitting while maximizing performance. Disabling the deep visual token setting degrades performance, largely because it slashes the trainable parameter count in the visual branch by roughly an order of magnitude ( $\approx 12\times$ ). With so few adaptive parameters, the frozen encoder cannot be steered adequately toward domain-specific features, so the overall accuracy suffers.

We find the best configuration that exhibited the most promising performance through a grid search, based on best accuracy and robustness in gastrointestinal pathology detection. Particularly, the combination of 64 class-specific context (use different text token for different class) and 5 deep visual token (use different visual token at every input layer of visual encoder) reaches the highest accuracy, indicating the significance of personalized context and deep visual representations in enhancing model performance.

Table 2 provides an extensive evaluation of TVTL-CLIP against representative state-of-the-art architectures spanning both convolutional and transformer families. We include DenseNet (four different depths configurations), RegNet, ResNet (three different depths configurations), lightweight SqueezeNet, the hierarchical Swin-v2, and four ViT variants, reporting each backbone in two

**Table 2: Comparison with SOTA architectures on the gastrointestinal lesion benchmark. "# params" reports trainable weights in millions. For each backbone we list results when trained from scratch and when finetuned from an ImageNet initialization (grey rows). All metrics are percentages; the proposed TVTL-CLIP achieves the best accuracy and specificity, requiring only 0.08M learnable parameters.**

| Model               | # params ( $\times 10^6$ ) | Acc.        | Recall      | Spec.       | F1 Score    |
|---------------------|----------------------------|-------------|-------------|-------------|-------------|
| DenseNet-121        | 6.96                       | 83.7        | 81.3        | 82.5        | 77.0        |
| ↪ Pre-trained       |                            | 88.7        | 84.1        | 86.0        | 79.0        |
| DenseNet-161        | 26.49                      | 85.9        | 83.6        | 82.9        | <b>81.9</b> |
| ↪ Pre-trained       |                            | 83.8        | 87.9        | 83.3        | 79.7        |
| DenseNet-169        | 12.50                      | 85.4        | 81.8        | 82.5        | 77.3        |
| ↪ Pre-trained       |                            | 83.7        | 91.1        | 80.5        | 77.9        |
| DenseNet-201        | 18.11                      | 85.5        | 82.2        | 80.5        | 81.4        |
| ↪ Pre-trained       |                            | 84.8        | 88.8        | 80.9        | 80.4        |
| RegNet              | 105.31                     | 70.3        | 57.5        | 73.5        | 51.6        |
| ↪ Pre-trained       |                            | 86.6        | 87.4        | 87.5        | 83.1        |
| ResNet-18           | 11.18                      | 78.6        | 84.6        | 71.2        | 72.6        |
| ↪ Pre-trained       |                            | 85.5        | 91.6        | 81.7        | 81.0        |
| ResNet-50           | 23.52                      | 73.5        | 78.0        | 66.1        | 63.0        |
| ↪ Pre-trained       |                            | 82.2        | <b>93.5</b> | 73.5        | 77.6        |
| ResNet-101          | 42.52                      | 79.8        | 74.3        | 72.4        | 69.3        |
| ↪ Pre-trained       |                            | 83.9        | 91.6        | 80.9        | 80.3        |
| SqueezeNet          | 0.73                       | 79.3        | 40.7        | 81.7        | 61.1        |
| ↪ Pre-trained       |                            | 85.6        | 75.2        | 86.4        | 76.1        |
| Swin Transformer v2 | 86.91                      | 55.3        | 86.0        | 21.4        | 18.8        |
| ↪ Pre-trained       |                            | 77.1        | 36.9        | 89.5        | 47.2        |
| ViT-B-16            | 85.80                      | 68.1        | 36.4        | 77.4        | 34.1        |
| ↪ Pre-trained       |                            | 72.6        | 48.6        | 79.8        | 49.4        |
| ViT-B-32            | 87.46                      | 76.5        | 38.8        | 80.9        | 50.6        |
| ↪ Pre-trained       |                            | 80.9        | 60.7        | 83.7        | 66.3        |
| ViT-L-16            | 303.52                     | 74.7        | 31.8        | 85.2        | 44.0        |
| ↪ Pre-trained       |                            | 76.5        | 45.3        | 85.6        | 47.5        |
| ViT-L-32            | 305.52                     | 82.6        | 36.4        | 90.3        | 56.6        |
| ↪ Pre-trained       |                            | 67.6        | 24.8        | 88.7        | 32.6        |
| <b>Ours</b>         | <b>0.08</b>                | <b>88.7</b> | <b>79.2</b> | <b>98.3</b> | <b>80.2</b> |

regimes: (i) trained from scratch on our endoscopic dataset and (ii) fine-tuned from ImageNet weights (grey rows). This dual view clarifies how strongly each model benefits from large-scale pretraining, and how much parameter overhead that benefit incurs.

Across the board, TVTL-CLIP delivers the highest accuracy (88.7%) and specificity (98.3%), while also posting competitive recall (79.2%) and F1 (80.2%). Especially, it achieves these scores with just 0.08M learned parameters, in stark contrast to the 6.9-305M range of the competing networks. DenseNet-161, the best conventional baseline for F1, requires 26.5M parameters (330x more than TVTL-CLIP) whereas the strongest ImageNet initialized CNNs and ViTs consume up to three orders of magnitude more capacity yet still trail our approach on at least two of the four metrics.

#### 4.4 Ablation Studies

To assess the individual contribution of each modality, we conducted systematic ablations by modifying the optimal configuration of TVTL-CLIP in three key ways: (i) retaining only textual prompts, (ii) retaining only visual prompts, and (iii) testing alternative CLIP backbones. Comparing these one-modality variants with the full model allows us to quantify the incremental value contributed by the learnable tokens injected into the text encoder versus those injected into the vision encoder for gastrointestinal-lesion classification.

In the *text-only* setup (Tab. 3), we removed all visual tokens and varied the number of learnable textual prompts from 0 to 69, testing both shared and class-specific context (CSC) settings. With the

**Table 3: Text-only ablation study. Performance is reported when no visual prompts are used. "Text Tok." denotes the number of learnable textual prompt tokens, while "CSC" marks whether class-specific context is applied or not. All metrics are expressed in %.**

| Text Tok.   | CSC | Acc.        | Recall      | Spec.       | F1 Score    |
|-------------|-----|-------------|-------------|-------------|-------------|
| 0           | -   | 48.2        | 10.6        | 85.8        | 10.2        |
| 1           | ✗   | 54.6        | 11.5        | 97.7        | 17.6        |
| 1           | ✓   | 54.6        | 11.5        | 97.7        | 17.6        |
| 2           | ✗   | 53.7        | 9.6         | 97.8        | 11.0        |
| 2           | ✓   | 53.7        | 9.6         | 97.8        | 11.0        |
| 4           | ✗   | 65.5        | 33.5        | 97.4        | 40.9        |
| 4           | ✓   | 65.5        | 33.5        | 97.4        | 40.9        |
| 8           | ✗   | 68.3        | 39.0        | 97.5        | 48.6        |
| 8           | ✓   | 68.5        | 39.5        | 97.5        | 49.2        |
| 16          | ✗   | 67.9        | 37.4        | 98.3        | 45.8        |
| 16          | ✓   | 67.9        | 37.4        | 98.3        | 45.8        |
| 32          | ✗   | 75.4        | 53.5        | 97.4        | 64.2        |
| 32          | ✓   | 75.4        | 53.5        | 97.4        | 64.2        |
| 64          | ✗   | 76.9        | 57.6        | 96.2        | 68.4        |
| 64          | ✓   | 76.9        | 57.6        | 96.2        | 68.4        |
| 69          | ✗   | 74.9        | 52.6        | 97.3        | 62.1        |
| 69          | ✓   | 76.1        | 55.7        | 96.5        | 63.1        |
| <b>Ours</b> | -   | <b>88.7</b> | <b>79.2</b> | <b>98.3</b> | <b>80.2</b> |

vision encoder frozen, the model struggles to adapt to the endoscopic domain, showing only modest improvements with increased prompt count (from 48.2% to 76.1% accuracy). CSC has negligible effect, with gains < 0.2 pp in all cases, indicating that when vision features cannot be modulated, textual prompting alone is insufficient to bridge the modality gap.

**Table 4: Visual-only ablation study. Performance is reported when no text prompts are used. "Vis. Tok." is the number of learnable visual prompt tokens; "Deep Vis. Tok." indicates whether those tokens are injected at every transformer block or only at the input layer. All metrics are in %.**

| Vis. Tok.   | Deep Vis. Tok. | Acc.        | Recall      | Spec.       | F1 Score    |
|-------------|----------------|-------------|-------------|-------------|-------------|
| 0           | -              | 48.2        | 10.6        | 85.8        | 10.2        |
| 1           | ✗              | 50.9        | 5.9         | 95.8        | 6.8         |
| 1           | ✓              | 65.4        | 33.5        | 97.3        | 40.1        |
| 3           | ✗              | 51.8        | 6.3         | 97.3        | 6.8         |
| 3           | ✓              | 76.4        | 55.2        | 97.6        | 59.1        |
| 5           | ✗              | 51.6        | 6.0         | 97.3        | 6.5         |
| 5           | ✓              | 84.2        | 70.8        | 97.6        | 77.9        |
| 6           | ✗              | 53.0        | 9.6         | 96.4        | 13.3        |
| 6           | ✓              | 80.1        | 62.3        | 97.9        | 69.1        |
| 7           | ✗              | 53.4        | 11.2        | 95.6        | 16.1        |
| 7           | ✓              | 84.6        | 71.4        | 97.9        | 73.6        |
| 8           | ✗              | 54.5        | 13.6        | 95.4        | 17.8        |
| 8           | ✓              | 81.8        | 66.0        | 97.6        | 66.4        |
| 9           | ✗              | 54.6        | 13.9        | 95.4        | 20.9        |
| 9           | ✓              | 83.6        | 69.4        | 97.8        | 74.3        |
| 10          | ✗              | 54.5        | 13.0        | 96.0        | 17.8        |
| 10          | ✓              | 82.8        | 68.0        | 97.5        | 72.8        |
| <b>Ours</b> | -              | <b>88.7</b> | <b>79.2</b> | <b>98.3</b> | <b>80.2</b> |

In the *visual-only* variant (Tab. 4), all text prompts were removed and 0-10 learnable visual tokens were injected, either at the input layer or recurrently at every transformer block (Deep Vis. Tok.). In this setting, the model displays much steeper gains: injecting even a single token per layer improves accuracy from 48.2% to 65.4%, and performance peaks at 84.6% with 7 tokens. Injecting tokens only at the input layer yields limited gains, while distributing them

across all layers ("deep prompting") significantly improves both recall and F1. However, performance saturates after 5-7 tokens, with additional prompts offering diminishing or even negative returns due to overfitting. The *full model* (last row in both tables), which combines 5 deep visual prompts and 64 CSC text tokens, achieves the best performance (Acc. 88.7%, Recall 79.2%, Spec. 98.3%, F1 80.2%), confirming that visual and textual modalities contribute complementary information.

**Table 5: Ablation on different CLIP visual backbone with best setup of TVTL-CLIP (Text Tok. = 64, CSC = True, Vis. Tok. = 5, Deep Vis. Tok. = Yes).**

| Visual Encoder Model   | Acc.        | Recall      | Spec.       | F1 Score    |
|------------------------|-------------|-------------|-------------|-------------|
| ResNet-50              | 87.8        | 78.5        | 96.9        | 76.3        |
| ResNet-50x4            | 85.4        | 72.3        | 98.4        | 68.6        |
| ResNet-50x16           | 86.4        | 72.2        | <b>99.0</b> | 74.9        |
| ResNet-50x64           | 87.7        | 78.6        | 96.2        | 76.9        |
| ResNet-101             | 82.8        | 68.1        | 98.7        | 68.2        |
| ViT-B-16               | 87.1        | 75.3        | 98.8        | 79.5        |
| <b>ViT-B-32 (Ours)</b> | <b>88.7</b> | <b>79.2</b> | <b>98.3</b> | <b>80.2</b> |

Finally, we ablated the CLIP visual backbone using this optimal prompt configuration (Tab. 5). Among all tested variants, ViT-B/32 delivers the best accuracy and F1 score, slightly outperforming both wider ResNet-based models and ViT-B/16. These results suggest that ViT-B/32 provides the best trade-off between representational capacity and efficiency in prompt-tuning scenarios.

Overall, the ablations confirm that (i) visual and textual prompts are individually beneficial but yield the highest accuracy when combined, (ii) deep visual prompting is substantially more effective than shallow injection or text-only tuning, and (iii) a small number of prompts per layer suffices, while larger sets lead to overfitting without consistent benefit.

## 5 Conclusion

We have presented TVTL-CLIP, a parameter-efficient framework that adapts a frozen CLIP model to gastrointestinal pathology detection by learning a small set of visual and textual prompt tokens. Extensive experiments on about 2k capsule endoscopy images show that TVTL-CLIP surpasses CNN and transformer baselines while training fewer than 0.1M parameters, confirming that joint prompt tuning can bridge the gap between generic vision-language pre-training and domain-specific medical tasks.

Ablation studies disentangle the roles of each modality: visual prompts alone provide strong embedding representations, textual prompts refine class discrimination, and their combination yields the highest accuracy, recall, specificity, and F1 score. These findings underscore the importance of balanced tokenization for capturing subtle visual patterns and specialized terminology.

Future work will investigate TVTL-CLIP in federated-learning scenarios to preserve patient privacy while aggregating knowledge across institutions, extend the approach to additional imaging modalities such as histopathology and radiology, and explore real-time deployment by optimizing inference on resource-constrained hardware. We believe the methodology outlined here offers a versatile path toward robust, data-efficient medical image analysis across a broad spectrum of clinical applications.

## CO<sub>2</sub> Emissions Related to Experiments

Experiments were conducted using a private infrastructure, with an estimated carbon efficiency of 0.432 kg CO<sub>2</sub>eq/kWh. A cumulative of 240 hours of computation was performed on 3 NVIDIA A100 GPUs (thermal design power of 300 W), with total emissions are estimated<sup>1</sup> to be 25.92 kg CO<sub>2</sub>eq.

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<sup>1</sup>Estimations were performed using the MachineLearning Impact calculator [10].

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